

Newman-Penrose Formalism Calculations

Stephani et al., Equation (17.4)

Metric

$$ds^2 = \epsilon e^M dt \otimes dt - \epsilon W e^\Psi dx \otimes dx - \epsilon A W e^\Psi dx \otimes dy - \epsilon A W e^\Psi dy \otimes dx - \left(\epsilon A^2 e^{(2\Psi)} - 1 \right) W e^{(-\Psi)} dy \otimes dy + e^M dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Metric Functions

$$\Psi(t, z) = \Psi$$

$$A(t, z) = A$$

$$W(t, z) = W$$

$$M(t, z) = M$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} \sqrt{\epsilon} \sqrt{W} e^{(\frac{1}{2}\Psi)} \right) dx + \left(\frac{1}{2} \sqrt{2} \sqrt{\epsilon} A \sqrt{W} e^{(\frac{1}{2}\Psi)} + \frac{1}{2} \sqrt{2} \sqrt{W} e^{(-\frac{1}{2}\Psi)} \right) dy$$

$$n = \left(\frac{1}{2} \sqrt{2} \sqrt{\epsilon} \sqrt{W} e^{(\frac{1}{2}\Psi)} \right) dx + \left(\frac{1}{2} \sqrt{2} \sqrt{\epsilon} A \sqrt{W} e^{(\frac{1}{2}\Psi)} - \frac{1}{2} \sqrt{2} \sqrt{W} e^{(-\frac{1}{2}\Psi)} \right) dy$$

$$m = \left(\frac{1}{2} \sqrt{2} \sqrt{\epsilon} e^{\left(\frac{1}{2} M\right)} \right) dt + \left(\frac{1}{2} i \sqrt{2} e^{\left(\frac{1}{2} M\right)} \right) dz$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} \sqrt{\epsilon} e^{\left(\frac{1}{2} M\right)} \right) dt + \left(-\frac{1}{2} i \sqrt{2} e^{\left(\frac{1}{2} M\right)} \right) dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2} A e^{\left(\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} - \frac{\sqrt{2} e^{\left(-\frac{1}{2} \Psi\right)}}{2 \sqrt{\epsilon} \sqrt{W}} \right) dx + \left(\frac{\sqrt{2} e^{\left(\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} \right) dy$$

$$n = \left(\frac{\sqrt{2} A e^{\left(\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} - \frac{\sqrt{2} e^{\left(-\frac{1}{2} \Psi\right)}}{2 \sqrt{\epsilon} \sqrt{W}} \right) dx + \left(-\frac{\sqrt{2} e^{\left(\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} \right) dy$$

$$m = \left(\frac{\sqrt{2} e^{\left(-\frac{1}{2} M\right)}}{2 \sqrt{\epsilon}} \right) dt + \left(\frac{1}{2} i \sqrt{2} e^{\left(-\frac{1}{2} M\right)} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2} e^{\left(-\frac{1}{2} M\right)}}{2 \sqrt{\epsilon}} \right) dt + \left(-\frac{1}{2} i \sqrt{2} e^{\left(-\frac{1}{2} M\right)} \right) dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2} A e^{\left(\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} - \frac{\sqrt{2} e^{\left(-\frac{1}{2} \Psi\right)}}{2 \sqrt{\epsilon} \sqrt{W}} \partial_x + \frac{\sqrt{2} e^{\left(\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} \partial_y$$

$$\Delta = \frac{\sqrt{2} A e^{\left(\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} - \frac{\sqrt{2} e^{\left(-\frac{1}{2} \Psi\right)}}{2 \sqrt{\epsilon} \sqrt{W}} \partial_x - \frac{\sqrt{2} e^{\left(\frac{1}{2} \Psi\right)}}{2 \sqrt{W}} \partial_y$$

$$\delta = \frac{\sqrt{2} e^{\left(-\frac{1}{2} M\right)}}{2 \sqrt{\epsilon}} \partial_t + \frac{1}{2} i \sqrt{2} e^{\left(-\frac{1}{2} M\right)} \partial_z$$

$$\bar{\delta} = \frac{\sqrt{2}e^{(-\frac{1}{2}M)}}{2\sqrt{\epsilon}}\partial_t - \frac{1}{2}i\sqrt{2}e^{(-\frac{1}{2}M)}\partial_z$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = \frac{1}{4}i\sqrt{2}\sqrt{\epsilon}e^{(-\frac{1}{2}M+\Psi)}\frac{\partial}{\partial z}A + \frac{1}{4}\sqrt{2}e^{(-\frac{1}{2}M+\Psi)}\frac{\partial}{\partial t}A - \frac{1}{4}i\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial z}\Psi - \frac{\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial t}\Psi}{4\sqrt{\epsilon}}$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = -\frac{1}{4}i\sqrt{2}\sqrt{\epsilon}e^{(-\frac{1}{2}M+\Psi)}\frac{\partial}{\partial z}A + \frac{1}{4}\sqrt{2}e^{(-\frac{1}{2}M+\Psi)}\frac{\partial}{\partial t}A - \frac{1}{4}i\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial z}\Psi + \frac{\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial t}\Psi}{4\sqrt{\epsilon}}$$

$$\alpha = -\frac{1}{8}i\sqrt{2}\sqrt{\epsilon}e^{(-\frac{1}{2}M+\Psi)}\frac{\partial}{\partial z}A + \frac{1}{8}\sqrt{2}e^{(-\frac{1}{2}M+\Psi)}\frac{\partial}{\partial t}A + \frac{1}{8}i\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial z}M - \frac{\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial t}M}{8\sqrt{\epsilon}}$$

$$\beta = \frac{1}{8}i\sqrt{2}\sqrt{\epsilon}e^{(-\frac{1}{2}M+\Psi)}\frac{\partial}{\partial z}A + \frac{1}{8}\sqrt{2}e^{(-\frac{1}{2}M+\Psi)}\frac{\partial}{\partial t}A + \frac{1}{8}i\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial z}M + \frac{\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial t}M}{8\sqrt{\epsilon}}$$

$$\tau = -\frac{i\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial z}W}{4W} - \frac{\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial t}W}{4\sqrt{\epsilon}W}$$

$$\pi = -\frac{i\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial z}W}{4W} + \frac{\sqrt{2}e^{(-\frac{1}{2}M)}\frac{\partial}{\partial t}W}{4\sqrt{\epsilon}W}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = & \frac{1}{4}\epsilon e^{(-M+2\Psi)}\frac{\partial}{\partial z}A^2 - \frac{1}{2}\sqrt{\epsilon}e^{(-M+\Psi)}\frac{\partial}{\partial z}A\frac{\partial}{\partial z}\Psi + \frac{1}{4}e^{(-M+2\Psi)}\frac{\partial}{\partial t}A^2 - \frac{1}{4}\sqrt{\epsilon}e^{(-M+\Psi)}\frac{\partial^2}{(\partial z)^2}A - \frac{e^{(-M+\Psi)}\frac{\partial}{\partial t}A\frac{\partial}{\partial t}\Psi}{2\sqrt{\epsilon}} - \frac{\sqrt{\epsilon}e^{(-M+\Psi)}\frac{\partial}{\partial z}A\frac{\partial}{\partial z}W}{4W} \\ & + \frac{1}{4}e^{(-M)}\frac{\partial^2}{(\partial z)^2}\Psi + \frac{e^{(-M)}\frac{\partial}{\partial z}\Psi\frac{\partial}{\partial z}W}{4W} - \frac{e^{(-M+\Psi)}\frac{\partial^2}{(\partial t)^2}A}{4\sqrt{\epsilon}} - \frac{e^{(-M+\Psi)}\frac{\partial}{\partial t}A\frac{\partial}{\partial t}W}{4\sqrt{\epsilon}W} + \frac{e^{(-M)}\frac{\partial^2}{(\partial t)^2}\Psi}{4\epsilon} + \frac{e^{(-M)}\frac{\partial}{\partial t}\Psi\frac{\partial}{\partial t}W}{4\epsilon W}\end{aligned}$$

$$\Phi_{01} = 0$$

$$\begin{aligned}\Phi_{02} = & -\frac{1}{8}\epsilon e^{(-M+2\Psi)}\frac{\partial}{\partial z}A^2 + \frac{1}{4}i\sqrt{\epsilon}e^{(-M+2\Psi)}\frac{\partial}{\partial t}A\frac{\partial}{\partial z}A + \frac{1}{8}e^{(-M+2\Psi)}\frac{\partial}{\partial t}A^2 + \frac{1}{8}e^{(-M)}\frac{\partial}{\partial z}\Psi^2 - \frac{ie^{(-M)}\frac{\partial}{\partial t}\Psi\frac{\partial}{\partial z}\Psi}{4\sqrt{\epsilon}} - \frac{e^{(-M)}\frac{\partial}{\partial t}\Psi^2}{8\epsilon} - \frac{e^{(-M)}\frac{\partial}{\partial z}M\frac{\partial}{\partial z}W}{4W} + \frac{ie^{(-M)}\frac{\partial}{\partial z}M\frac{\partial}{\partial t}W}{4\sqrt{\epsilon}W} \\ & + \frac{ie^{(-M)}\frac{\partial}{\partial t}M\frac{\partial}{\partial z}W}{4\sqrt{\epsilon}W} + \frac{e^{(-M)}\frac{\partial}{\partial t}M\frac{\partial}{\partial t}W}{4\epsilon W} - \frac{e^{(-M)}\frac{\partial}{\partial z}W^2}{8W^2} + \frac{e^{(-M)}\frac{\partial^2}{(\partial z)^2}W}{4W} - \frac{ie^{(-M)}\frac{\partial^2}{\partial t\partial z}W}{2\sqrt{\epsilon}W} + \frac{ie^{(-M)}\frac{\partial}{\partial t}W\frac{\partial}{\partial z}W}{4\sqrt{\epsilon}W^2} + \frac{e^{(-M)}\frac{\partial}{\partial t}W^2}{8\epsilon W^2} - \frac{e^{(-M)}\frac{\partial^2}{(\partial t)^2}W}{4\epsilon W}\end{aligned}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{1}{16}\epsilon e^{(-M+2\Psi)}\frac{\partial}{\partial z}A^2 + \frac{1}{16}e^{(-M+2\Psi)}\frac{\partial}{\partial t}A^2 - \frac{1}{16}e^{(-M)}\frac{\partial}{\partial z}\Psi^2 - \frac{1}{8}e^{(-M)}\frac{\partial^2}{(\partial z)^2}M - \frac{e^{(-M)}\frac{\partial}{\partial t}\Psi^2}{16\epsilon} - \frac{e^{(-M)}\frac{\partial^2}{(\partial t)^2}M}{8\epsilon} + \frac{e^{(-M)}\frac{\partial}{\partial z}W^2}{16W^2} + \frac{e^{(-M)}\frac{\partial}{\partial t}W^2}{16\epsilon W^2}$$

$$\Phi_{12} = 0$$

$$\begin{aligned}\Phi_{20} = & -\frac{1}{8}\epsilon e^{(-M+2\Psi)}\frac{\partial}{\partial z}A^2 - \frac{1}{4}i\sqrt{\epsilon}e^{(-M+2\Psi)}\frac{\partial}{\partial t}A\frac{\partial}{\partial z}A + \frac{1}{8}e^{(-M+2\Psi)}\frac{\partial}{\partial t}A^2 + \frac{1}{8}e^{(-M)}\frac{\partial}{\partial z}\Psi^2 + \frac{ie^{(-M)}\frac{\partial}{\partial t}\Psi\frac{\partial}{\partial z}\Psi}{4\sqrt{\epsilon}} - \frac{e^{(-M)}\frac{\partial}{\partial t}\Psi^2}{8\epsilon} - \frac{e^{(-M)}\frac{\partial}{\partial z}M\frac{\partial}{\partial z}W}{4W} - \frac{ie^{(-M)}\frac{\partial}{\partial z}M\frac{\partial}{\partial t}W}{4\sqrt{\epsilon}W} \\ & - \frac{ie^{(-M)}\frac{\partial}{\partial t}M\frac{\partial}{\partial z}W}{4\sqrt{\epsilon}W} + \frac{e^{(-M)}\frac{\partial}{\partial t}M\frac{\partial}{\partial t}W}{4\epsilon W} - \frac{e^{(-M)}\frac{\partial}{\partial z}W^2}{8W^2} + \frac{e^{(-M)}\frac{\partial^2}{(\partial z)^2}W}{4W} + \frac{ie^{(-M)}\frac{\partial^2}{\partial t\partial z}W}{2\sqrt{\epsilon}W} - \frac{ie^{(-M)}\frac{\partial}{\partial t}W\frac{\partial}{\partial z}W}{4\sqrt{\epsilon}W^2} + \frac{e^{(-M)}\frac{\partial}{\partial t}W^2}{8\epsilon W^2} - \frac{e^{(-M)}\frac{\partial^2}{(\partial t)^2}W}{4\epsilon W}\end{aligned}$$

$$\Phi_{21} = 0$$

$$\begin{aligned}\Phi_{22} = & \frac{1}{4} \epsilon e^{(-M+2\Psi)} \frac{\partial}{\partial z} A^2 + \frac{1}{2} \sqrt{\epsilon} e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} \Psi + \frac{1}{4} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A^2 + \frac{1}{4} \sqrt{\epsilon} e^{(-M+\Psi)} \frac{\partial^2}{(\partial z)^2} A + \frac{e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} \Psi}{2 \sqrt{\epsilon}} + \frac{\sqrt{\epsilon} e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} W}{4 W} \\ & + \frac{1}{4} e^{(-M)} \frac{\partial^2}{(\partial z)^2} \Psi + \frac{e^{(-M)} \frac{\partial}{\partial z} \Psi \frac{\partial}{\partial z} W}{4 W} + \frac{e^{(-M+\Psi)} \frac{\partial^2}{(\partial t)^2} A}{4 \sqrt{\epsilon}} + \frac{e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} W}{4 \sqrt{\epsilon} W} + \frac{e^{(-M)} \frac{\partial^2}{(\partial t)^2} \Psi}{4 \epsilon} + \frac{e^{(-M)} \frac{\partial}{\partial t} \Psi \frac{\partial}{\partial t} W}{4 \epsilon W}\end{aligned}$$

$$\Lambda = \frac{1}{48} \epsilon e^{(-M+2\Psi)} \frac{\partial}{\partial z} A^2 + \frac{1}{48} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A^2 - \frac{1}{48} e^{(-M)} \frac{\partial}{\partial z} \Psi^2 - \frac{1}{24} e^{(-M)} \frac{\partial^2}{(\partial z)^2} M - \frac{e^{(-M)} \frac{\partial}{\partial t} \Psi^2}{48 \epsilon} - \frac{e^{(-M)} \frac{\partial^2}{(\partial t)^2} M}{24 \epsilon} + \frac{e^{(-M)} \frac{\partial}{\partial z} W^2}{48 W^2} - \frac{e^{(-M)} \frac{\partial^2}{(\partial z)^2} W}{12 W} + \frac{e^{(-M)} \frac{\partial}{\partial t} W^2}{48 \epsilon W^2} - \frac{e^{(-M)} \frac{\partial^2}{(\partial t)^2} W}{12 \epsilon W}$$

Weyl Tensor Components

$$\begin{aligned}\Psi_0 = & -\frac{1}{4} \epsilon e^{(-M+2\Psi)} \frac{\partial}{\partial z} A^2 + \frac{1}{2} i \sqrt{\epsilon} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} A - \frac{1}{4} \sqrt{\epsilon} e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} M + \frac{1}{2} \sqrt{\epsilon} e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} \Psi + \frac{1}{4} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A^2 + \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial t} M \\ & + \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} M - \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial t} \Psi - \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} \Psi + \frac{1}{4} e^{(-M)} \frac{\partial}{\partial z} M \frac{\partial}{\partial z} \Psi + \frac{1}{4} \sqrt{\epsilon} e^{(-M+\Psi)} \frac{\partial^2}{(\partial z)^2} A + \frac{e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} M}{4 \sqrt{\epsilon}} - \frac{e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} \Psi}{2 \sqrt{\epsilon}} \\ & - \frac{i e^{(-M)} \frac{\partial}{\partial z} M \frac{\partial}{\partial t} \Psi}{4 \sqrt{\epsilon}} - \frac{i e^{(-M)} \frac{\partial}{\partial t} M \frac{\partial}{\partial z} \Psi}{4 \sqrt{\epsilon}} + \frac{\sqrt{\epsilon} e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial z} W}{4 W} - \frac{1}{2} i e^{(-M+\Psi)} \frac{\partial^2}{\partial t \partial z} A - \frac{e^{(-M)} \frac{\partial}{\partial t} M \frac{\partial}{\partial t} \Psi}{4 \epsilon} - \frac{1}{4} e^{(-M)} \frac{\partial^2}{(\partial z)^2} \Psi - \frac{i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial t} W}{4 W} - \frac{i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} W}{4 W} \\ & - \frac{e^{(-M)} \frac{\partial}{\partial z} \Psi \frac{\partial}{\partial z} W}{4 W} - \frac{e^{(-M+\Psi)} \frac{\partial^2}{(\partial t)^2} A}{4 \sqrt{\epsilon}} + \frac{i e^{(-M)} \frac{\partial^2}{\partial t \partial z} \Psi}{2 \sqrt{\epsilon}} - \frac{e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial t} W}{4 \sqrt{\epsilon} W} + \frac{i e^{(-M)} \frac{\partial}{\partial z} \Psi \frac{\partial}{\partial t} W}{4 \sqrt{\epsilon} W} + \frac{i e^{(-M)} \frac{\partial}{\partial t} \Psi \frac{\partial}{\partial z} W}{4 \sqrt{\epsilon} W} + \frac{e^{(-M)} \frac{\partial^2}{(\partial t)^2} \Psi}{4 \epsilon} + \frac{e^{(-M)} \frac{\partial}{\partial t} \Psi \frac{\partial}{\partial t} W}{4 \epsilon W}\end{aligned}$$

$$\Psi_1 = 0$$

$$\begin{aligned}\Psi_2 = & \frac{1}{12} \epsilon e^{(-M+2\Psi)} \frac{\partial}{\partial z} A^2 + \frac{1}{12} e^{(-M+2\Psi)} \frac{\partial}{\partial t} A^2 + \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial}{\partial z} A \frac{\partial}{\partial t} \Psi - \frac{1}{4} i e^{(-M+\Psi)} \frac{\partial}{\partial t} A \frac{\partial}{\partial z} \Psi - \frac{1}{12} e^{(-M)} \frac{\partial}{\partial z} \Psi^2 \\ & + \frac{1}{12} e^{(-M)} \frac{\partial^2}{(\partial z)^2} M - \frac{e^{(-M)} \frac{\partial}{\partial t} \Psi^2}{12 \epsilon} + \frac{e^{(-M)} \frac{\partial^2}{(\partial t)^2} M}{12 \epsilon} + \frac{e^{(-M)} \frac{\partial}{\partial z} W^2}{12 W^2} - \frac{e^{(-M)} \frac{\partial^2}{(\partial z)^2} W}{12 W} + \frac{e^{(-M)} \frac{\partial}{\partial t} W^2}{12 \epsilon W^2} - \frac{e^{(-M)} \frac{\partial^2}{(\partial t)^2} W}{12 \epsilon W}\end{aligned}$$

$$\Psi_3 = 0$$

$$\begin{aligned}
\Psi_4 = & -\frac{1}{4}\epsilon e^{(-M+2\Psi)}\frac{\partial}{\partial z}A^2 - \frac{1}{2}i\sqrt{\epsilon}e^{(-M+2\Psi)}\frac{\partial}{\partial t}A\frac{\partial}{\partial z}A + \frac{1}{4}\sqrt{\epsilon}e^{(-M+\Psi)}\frac{\partial}{\partial z}A\frac{\partial}{\partial z}M - \frac{1}{2}\sqrt{\epsilon}e^{(-M+\Psi)}\frac{\partial}{\partial z}A\frac{\partial}{\partial z}\Psi + \frac{1}{4}e^{(-M+2\Psi)}\frac{\partial}{\partial t}A^2 + \frac{1}{4}i e^{(-M+\Psi)}\frac{\partial}{\partial z}A\frac{\partial}{\partial t}M \\
& + \frac{1}{4}i e^{(-M+\Psi)}\frac{\partial}{\partial t}A\frac{\partial}{\partial z}M - \frac{1}{2}i e^{(-M+\Psi)}\frac{\partial}{\partial z}A\frac{\partial}{\partial t}\Psi - \frac{1}{2}i e^{(-M+\Psi)}\frac{\partial}{\partial t}A\frac{\partial}{\partial z}\Psi + \frac{1}{4}e^{(-M)}\frac{\partial}{\partial z}M\frac{\partial}{\partial z}\Psi - \frac{1}{4}\sqrt{\epsilon}e^{(-M+\Psi)}\frac{\partial^2}{(\partial z)^2}A - \frac{e^{(-M+\Psi)}\frac{\partial}{\partial t}A\frac{\partial}{\partial t}M}{4\sqrt{\epsilon}} + \frac{e^{(-M+\Psi)}\frac{\partial}{\partial t}A\frac{\partial}{\partial t}\Psi}{2\sqrt{\epsilon}} \\
& + \frac{i e^{(-M)}\frac{\partial}{\partial z}M\frac{\partial}{\partial t}\Psi}{4\sqrt{\epsilon}} + \frac{i e^{(-M)}\frac{\partial}{\partial t}M\frac{\partial}{\partial z}\Psi}{4\sqrt{\epsilon}} - \frac{\sqrt{\epsilon}e^{(-M+\Psi)}\frac{\partial}{\partial z}A\frac{\partial}{\partial z}W}{4W} - \frac{1}{2}i e^{(-M+\Psi)}\frac{\partial^2}{\partial t\partial z}A - \frac{e^{(-M)}\frac{\partial}{\partial t}M\frac{\partial}{\partial t}\Psi}{4\epsilon} - \frac{1}{4}e^{(-M)}\frac{\partial^2}{(\partial z)^2}\Psi - \frac{i e^{(-M+\Psi)}\frac{\partial}{\partial z}A\frac{\partial}{\partial t}W}{4W} - \frac{i e^{(-M+\Psi)}\frac{\partial}{\partial t}A\frac{\partial}{\partial z}W}{4W} \\
& - \frac{e^{(-M)}\frac{\partial}{\partial z}\Psi\frac{\partial}{\partial z}W}{4W} + \frac{e^{(-M+\Psi)}\frac{\partial^2}{(\partial t)^2}A}{4\sqrt{\epsilon}} - \frac{i e^{(-M)}\frac{\partial^2}{\partial t\partial z}\Psi}{2\sqrt{\epsilon}} + \frac{e^{(-M+\Psi)}\frac{\partial}{\partial t}A\frac{\partial}{\partial t}W}{4\sqrt{\epsilon}W} - \frac{i e^{(-M)}\frac{\partial}{\partial z}\Psi\frac{\partial}{\partial t}W}{4\sqrt{\epsilon}W} - \frac{i e^{(-M)}\frac{\partial}{\partial t}\Psi\frac{\partial}{\partial z}W}{4\sqrt{\epsilon}W} + \frac{e^{(-M)}\frac{\partial^2}{(\partial t)^2}\Psi}{4\epsilon} + \frac{e^{(-M)}\frac{\partial}{\partial t}\Psi\frac{\partial}{\partial t}W}{4\epsilon W}
\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"